1475. Final Prices With a Special Discount in a Shop

Solved

Easy | Topics | Companies | Hint

You are given an integer array `prices` where `prices[i]` is the price of the i^{th} item in a shop.

There is a special discount for items in the shop. If you buy the i^{th} item, then you will receive a discount equivalent to `prices[j]` where `j` is the minimum index such that `j > i` and `prices[j] <= prices[i]`. Otherwise, you will not receive any discount at all.

Return an integer array `answer` where `answer[i]` is the final price you will pay for the i^{th} item of the shop, considering the special discount.

Example 1:

```
Input: prices = [8,4,6,2,3]
Output: [4,2,4,2,3]
Explanation:
For item 0 with price[0]=8 you will receive a discount equivalent to prices[1]=4, therefore, the final price you will pay is 8 - 4 = 4.
For item 1 with price[1]=4 you will receive a discount equivalent to prices[3]=2, therefore, the final price you will pay is 4 - 2 = 2.
For item 2 with price[2]=6 you will receive a discount equivalent to prices[3]=2, therefore, the final price you will pay is 6 - 2 = 4.
For items 3 and 4 you will not receive any discount at all.
```

Example 2:

```
Input: prices = [1,2,3,4,5]
Output: [1,2,3,4,5]
Explanation: In this case, for all items, you will not receive any discount at all.
```

Example 3:

```
Input: prices = [10,1,1,6]
Output: [9,0,1,6]
```

Constraints:

- `1 <= prices.length <= 500`
- `1 <= prices[i] <= 1000`

```java
class Solution {
    public int[] finalPrices(int[] prices) {
        int result[] = new int[prices.length];
        Deque<Integer> stack = new ArrayDeque<>();
        for (int i = 0; i < prices.length; i++) {
            while (!stack.isEmpty() && prices[stack.peek()] >= prices[i]) {
                result[stack.pop()] = prices[i];
            }
            stack.push(i);
        }
        for (int i = 0; i < prices.length; i++) {
```

```
            result[i] = prices[i] - result[i];
        }
        return result;

    }
}
```